

Agent Based Modeling for Group Alignment

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September 2026

1 Context

As AI agents will outnumber humans on the Internet very soon (citation needed), it is important to align group agent behaviours to ensure agents conform to values accepted by humans.

Given that individually aligned agents become misaligned when working with other agents (citation needed from OpenAI–HuggingFace incident), it is therefore important to apply group alignment on a global level.

Our hypotheses are:

- Fast agents mainly interact with each other
- Information from fast agents rapidly propagates throughout the collective and drowns out information from slow agents
- When fast agents are encouraged to consult slow agents more often, information from slow agents can propagate and compete with information from fast agents.

2 Notes

Key assumptions and definitions:

- Mixed swarms can be modelled by simple stochastic agents
- Stochastic agents can be modelled as a simple mathematical function $P(\text{action} \mid \text{state})$
- Every agent has a fixed *type* (fast or slow) and a mutable *state*, which is the origin label of the information it currently holds: F (fast-agent origin) or S (slow-agent origin).
- Every agent is seeded with its own type's label: fast agents start in state F , slow agents start in state S . The state is binary and switching is allowed in both directions (a two-species voter/invasion process).
- An agent has only one simplistic action: adopt and propagate information from neighbouring agents given their state.
- Adoption and propagation occur at the same time. Adoption is defined as internalising the information, while propagation means sharing that internalised information with neighbours. An agent propagates the information as soon as it adopts it. The dynamics are *push* on the propagator's side and *pull* on the adopter's side: when an agent activates it offers its current label to every neighbour, and each neighbour independently decides whether to take it up. The adopter's decision is what the probabilities P_{ab} below describe.
- Conclusions from simple stochastic agents are hypothesised to scale to mixed swarms
- AI agents are much faster than humans at acquiring and propagating information (e.g. fast agent = AI agent, slow agent = human). The reason is that humans react much more slowly to information than AI agents.
- The topology is an $n \times n$ grid with boundaries (not a torus). Agents on an edge or in a corner simply have fewer neighbours (5 or 3 instead of 8). n is taken large enough that boundary agents are a small fraction of the population and edge effects are negligible.

3 Experimental methodology

3.1 Topology

Each agent/human is able to receive and send information to others. Assume that agents and humans are randomly distributed on an $n \times n$ 2D grid with a Moore neighbourhood, so that each interior agent has $k = 8$ neighbours as shown in Figure 1; agents on the boundary have fewer. The Moore neighbourhood is preferred over the 4-neighbour von Neumann neighbourhood because actual agent–human interaction networks are far more highly connected than a nearest-neighbour lattice (citation needed); the grid is a deliberate simplification of a real small-world network.

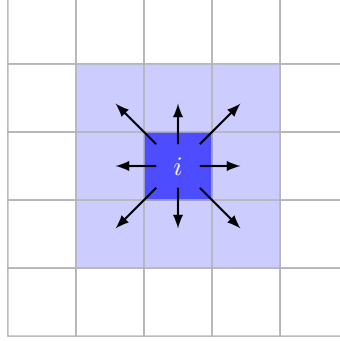


Figure 1: Moore neighbourhood: an interior agent i propagates to its $k = 8$ surrounding cells (shaded). On the $n \times n$ grid, edge and corner agents have 5 and 3 neighbours respectively.

3.2 Timing

Each fast agent is set for a sweep of

$$\rho \equiv \frac{\Delta T_s}{\Delta T_f} \in \{1, 3, 10, 100\}$$

times faster than a slow agent to process information, where $\rho = 1$ is the base case in which all agent types are treated the same in the adopt–propagate cycle. ΔT_f is the time interval for a fast agent to adopt and propagate information, while ΔT_s is the time interval for a slow agent to adopt and propagate information. Equivalently, fast and slow agents activate at rates

$$r_f = \frac{1}{\Delta T_f}, \quad r_s = \frac{1}{\Delta T_s} = \frac{r_f}{\rho}.$$

In simulation, activation is modelled as an independent Poisson clock per agent (continuous time) or as random asynchronous updates in which a fast agent is chosen ρ times as often as a slow agent (discrete time); the two are equivalent for the quantities measured here.

3.3 Adoption probabilities

Each agent keeps on iteratively adopting and propagating information stochastically depending on the probability of acceptance, which we identify as the key variables that should be controlled in our third hypothesis.

Let P_{ab} denote the probability that an agent of type b adopts what a neighbouring *propagator* of type a offers, for $a, b \in \{f, s\}$. Thus P_{ff} is the probability that a fast agent adopts what a fast neighbour offers, P_{sf} is the probability that a fast agent adopts what a slow neighbour offers, P_{fs} is the probability that a slow agent adopts what a fast neighbour offers, and P_{ss} the probability that a slow agent adopts what a slow neighbour offers. In total we have four variables that can be controlled: P_{ff} , P_{sf} , P_{fs} and P_{ss} . Crucially, the probability of a state change is independent of whether the information *originated* from a fast or slow agent; it depends only on whether the *propagator* is fast or slow. Agents do not know the provenance of what they are offered, only who is offering it.

3.4 Formal model

Let there be N agents, a fraction f of which are fast ($N_f = fN$, $N_s = (1 - f)N$). Agent i has type $\tau(i) \in \{f, s\}$ and state $x_i(t) \in \{F, S\}$, with $x_i(0) = F$ if $\tau(i) = f$ and $x_i(0) = S$ otherwise. The dynamics are:

1. Agent i activates at rate $r_{\tau(i)}$.
2. On activation, i offers x_i to each of its neighbours j (8 for an interior agent).
3. Each neighbour j independently sets $x_j \leftarrow x_i$ with probability $P_{\tau(i)\tau(j)}$; otherwise x_j is unchanged.

Hence the per-agent stochastic policy is

$$P(\text{adopt} \mid \text{offered by type } a, \text{ own type } b) = P_{ab},$$

and a single directed edge $i \rightarrow j$ carries successful transmissions at rate $r_{\tau(i)} P_{\tau(i)\tau(j)}$.

3.5 Outcome measure: the $\mathcal{C}$ -index

Let $n_F(t)$ and $n_S(t)$ be the number of agents holding label F and S at time t . The resulting index is evaluated at a long horizon T (in expectation over independent runs) and measures the proportion of slow-origin information relative to the proportion of slow agents in the initial population:

$$\mathcal{C} = \frac{\mathbb{E}[n_S(T)] / \mathbb{E}[n_F(T)]}{N_s / N_f} = \frac{\text{slow-origin information} / \text{fast-origin information}}{\text{number of slow agents} / \text{number of fast agents}}.$$

If $\mathcal{C} < 1$, fast-origin information outcompetes slow-origin information relative to the population mix; if $\mathcal{C} = 1$, each type's information is represented in proportion to its numbers; if $\mathcal{C} > 1$, slow-origin information outcompetes fast-origin information.

On a finite grid the process eventually reaches consensus (one label fixates), so a single run at very large T gives $n_S \in \{0, N\}$. The expectation over runs is therefore governed by the *fixation probability* of each label, and $\mathcal{C}$ is equivalently the fixation odds $\Pr[S \text{ fixates}] / \Pr[F \text{ fixates}]$ normalised by N_s / N_f . Both readings coincide (Proposition 1), so in practice $\mathcal{C}$ can be estimated either from label fractions at a fixed horizon or from fixation counts over many seeds.

4 Hypotheses testing

For the purpose of evaluating the first two hypotheses, let the probability of an agent state change be independent of the identity of its neighbour (fast or slow), so without loss of generality, let $P_{ff} = P_{sf} = P_{fs} = P_{ss} = p = 0.1$. Unless stated otherwise the population is balanced, $f = \frac{1}{2}$, so that any asymmetry in the results is attributable to speed rather than numbers.

4.1 Hypothesis 1: fast agents mainly interact with each other

Since agents are fixed on the grid, “interact” is operationalised as a *successful adoption event*: one agent offers its label, and a neighbour accepts it. The hypothesis holds because fast agents take more turns, so most offers come from fast agents, and about half of those offers land on other fast agents.

Concretely, the mathematical modelling is a counting argument. Take a balanced population of $N_f = N_s = N/2$ fast and slow agents, and count adoption events over one slow time step ΔT_s . In that window every slow agent takes exactly one turn and every fast agent takes ρ turns. On each turn an agent offers its label to its 8 neighbours; because agents are placed at random, on average 4 of those neighbours are fast and 4 are slow. Each offer is accepted with probability p . Multiplying

$$(\text{number of agents}) \times (\text{turns each}) \times (\text{neighbours of the receiving type}) \times (\text{acceptance probability})$$

gives the expected number of adoption events of each kind:

$$\begin{aligned} \text{fast} \rightarrow \text{fast} &: \frac{N}{2} \times \rho \times 4 \times p = 2\rho Np, \\ \text{fast} \rightarrow \text{slow} &: \frac{N}{2} \times \rho \times 4 \times p = 2\rho Np, \\ \text{slow} \rightarrow \text{fast} &: \frac{N}{2} \times 1 \times 4 \times p = 2Np, \\ \text{slow} \rightarrow \text{slow} &: \frac{N}{2} \times 1 \times 4 \times p = 2Np. \end{aligned}$$

The total is $4Np(\rho + 1)$. Dividing, the share of all adoption events that are fast→fast is

$$\sigma_{ff} = \frac{2\rho Np}{4Np(\rho + 1)} = \frac{\rho}{2(\rho + 1)}, \quad (1)$$

and the share of events *started* by a fast agent is

$$\sigma_{f\cdot} = \frac{2\rho Np + 2\rho Np}{4Np(\rho + 1)} = \frac{\rho}{\rho + 1}. \quad (2)$$

Both N and p cancel, so the result depends only on the speed ratio. When $\rho = 1$ the four kinds of event are equally common ($\sigma_{ff} = \frac{1}{4}$), which is exactly what random mixing predicts. As ρ grows, $\sigma_{f\cdot} \rightarrow 1$ and $\sigma_{ff} \rightarrow \frac{1}{2}$: almost every interaction is started by a fast agent, and half of all interactions are fast agents talking to other fast agents. Notice, however, that the share of events *received* by fast agents is $(2\rho Np + 2Np)/4Np(\rho + 1) = \frac{1}{2}$ for every ρ . Fast agents do not receive more than their share; they *send* more than their share.

Worked example: with 100 fast and 100 slow agents, $\rho = 3$ and $p = 0.1$, one slow time step produces on average 120 fast→fast, 120 fast→slow, 40 slow→fast and 40 slow→slow adoptions (320 in total). Three quarters of all interactions are started by a fast agent, and slow agents start only one quarter.

Boundary agents on the $n \times n$ grid have fewer than 8 neighbours, but this scales all four counts by the same factor and leaves the shares unchanged. For an unbalanced population with a fraction f of fast agents, the same counting gives $\sigma_{ff} = f^2/(f + (1 - f)/\rho)$ and $\sigma_{f\cdot} = f/(f + (1 - f)/\rho)$, which reduce to the expressions above at $f = \frac{1}{2}$. The empirical test is the 2×2 table of adoption-event counts, compared against the equal-shares baseline of $\frac{1}{4}$ each.

The results from the experiment show that

[Placeholder – results pending]

4.2 Hypothesis 2: fast-origin information drowns out slow-origin information

The second hypothesis is that information from fast agents rapidly propagates throughout the collective and drowns out information from slow agents. The reason is that fast agents produce more information overall, so gradually information that originates from fast agents dominates.

Concretely, the mathematical modelling shows the following. Let $u_f(t)$ and $u_s(t)$ be the fraction of fast and slow agents, respectively, holding label F at time t , with $u_f(0) = 1$ and $u_s(0) = 0$. In a mean-field (well-mixed) approximation, writing $\bar{k}$ for the average number of neighbours on the $n \times n$ grid ($\bar{k} \rightarrow 8$ as n grows), an agent of type b receives label- F offers at rate $\bar{k}[fr_f P_{fb} u_f + (1 - f)r_s P_{sb} u_s]$ and label- S offers at rate $\bar{k}[fr_f P_{fb}(1 - u_f) + (1 - f)r_s P_{sb}(1 - u_s)]$. Balancing gains and losses gives

$$\frac{du_b}{dt} = \bar{k} \sum_{a \in \{f, s\}} w_{ab} (u_a - u_b), \quad w_{ab} \equiv f_a r_a P_{ab}, \quad (3)$$

which is a linear consensus (DeGroot) system: the weight w_{ab} is the influence of type a on type b . For two types the within-type terms vanish and

$$\frac{du_f}{dt} = \bar{k} w_{sf} (u_s - u_f), \quad \frac{du_s}{dt} = \bar{k} w_{fs} (u_f - u_s). \quad (4)$$

The difference $u_f - u_s$ decays as $\exp[-\bar{k}(w_{fs} + w_{sf})t]$, so both types converge to a common value on a timescale

$$\tau = \frac{1}{\bar{k}(w_{fs} + w_{sf})} = \frac{\Delta T_f}{\bar{k}p(f + (1-f)/\rho)} \xrightarrow{\rho \rightarrow \infty} \frac{\Delta T_f}{\bar{k}pf}, \quad (5)$$

i.e. consensus is reached on the *fast* agents' clock regardless of how slow the slow agents are. This is the “rapidly” in the hypothesis. The quantity $w_{fs}u_f + w_{sf}u_s$ is conserved by (4), so the common limit is

$$u^* = \frac{w_{fs}u_f(0) + w_{sf}u_s(0)}{w_{fs} + w_{sf}} = \frac{f r_f P_{fs}}{f r_f P_{fs} + (1-f) r_s P_{sf}}. \quad (6)$$

With uniform p and $r_f = \rho r_s$ this reduces to

$$u^* = \frac{f\rho}{f\rho + (1-f)}, \quad 1 - u^* = \frac{1-f}{f\rho + (1-f)}, \quad (7)$$

so for $f = \frac{1}{2}$ the fast-origin label ends up held by a fraction $\rho/(\rho+1)$ of the population: 75% at $\rho = 3$, 91% at $\rho = 10$, 99% at $\rho = 100$.

The mean-field limit is not merely approximate here; the fixation probability it predicts is exact on any topology.

Proposition 1 (Fixation probability). *Let $c_f = r_f P_{fs}$ and $c_s = r_s P_{sf}$, and define*

$$M(t) = c_f \sum_{i: \tau(i)=f} \mathbf{1}[x_i(t) = F] + c_s \sum_{i: \tau(i)=s} \mathbf{1}[x_i(t) = F].$$

Then $M(t)$ is a martingale for the dynamics of Section 3.4 on any finite graph, including the bounded $n \times n$ grid (the argument never uses the number of neighbours). Consequently

$$\Pr[F \text{ fixates}] = \frac{M(0)}{M_{\max}} = \frac{N_f c_f}{N_f c_f + N_s c_s} = u^*, \quad \Pr[S \text{ fixates}] = 1 - u^*. \quad (8)$$

Proof sketch. Consider a single edge $\{i, j\}$. If both endpoints have the same type a , transmissions $i \rightarrow j$ and $j \rightarrow i$ each occur at rate $r_a P_{aa}$ and change M by $\pm c_a(\mathbf{1}[x_i = F] - \mathbf{1}[x_j = F])$ with opposite signs, so the expected change is zero. If $\tau(i) = f$ and $\tau(j) = s$, the transmission $i \rightarrow j$ occurs at rate $r_f P_{fs}$ and changes M by $c_s(\mathbf{1}[x_i = F] - \mathbf{1}[x_j = F])$; the transmission $j \rightarrow i$ occurs at rate $r_s P_{sf}$ and changes M by $c_f(\mathbf{1}[x_j = F] - \mathbf{1}[x_i = F])$. The expected drift along the edge is proportional to $r_f P_{fs} c_s - r_s P_{sf} c_f = 0$ by the choice of c_f, c_s . Summing over edges, $\mathbb{E}[dM] = 0$. Since the process is absorbed at $M = 0$ (all S) or $M = M_{\max} = N_f c_f + N_s c_s$ (all F), the optional stopping theorem gives $\Pr[F \text{ fixates}] = M(0)/M_{\max}$ with $M(0) = N_f c_f$. $\square$

Substituting into the definition of the $\mathcal{C}$ -index,

$$\mathcal{C} = \frac{\Pr[S \text{ fixates}]/\Pr[F \text{ fixates}]}{N_s/N_f} = \frac{c_s}{c_f} = \frac{r_s P_{sf}}{r_f P_{fs}} = \frac{P_{sf}}{\rho P_{fs}}. \quad (9)$$

Under the uniform-probability setting of Hypotheses 1 and 2 this gives

$$\mathcal{C} = \frac{1}{\rho}, \quad (10)$$

independently of f , N , the grid size and shape, and the within-type probabilities P_{ff}, P_{ss} . Every doubling of the speed ratio halves the representation of slow-origin information. Table 1 collects the predictions for the sweep.

Remark 1. *Spatial correlations on the grid (domains of like-labelled agents) affect the time to fixation, which is known to be slow for two-dimensional voter dynamics, but not the fixation probability, which is fixed by Proposition 1. Simulations should therefore report both: $\mathcal{C}$ estimated from fixation counts (or from label fractions at horizon $T \gg \tau$), and the empirical distribution of fixation times.*

The results from the experiment show that

[Placeholder – results pending]

Table 1: Analytic predictions for $f = \frac{1}{2}$, uniform $p = 0.1$.

ρ	fast→fast share σ_{ff}	fast-initiated share σ_f	$\Pr[S \text{ fixates}]$	$\mathcal{C}$
1	0.250	0.500	0.500	1.00
3	0.375	0.750	0.250	0.33
10	0.455	0.909	0.091	0.10
100	0.495	0.990	0.010	0.01

4.3 Hypothesis 3: consultation restores balance

The third hypothesis is that when fast agents are encouraged to consult slow agents more often, information from slow agents can propagate and compete with information from fast agents. The group alignment intervention is done by raising P_{sf} , so that slow-origin information benefits indirectly because slow agents mostly carry slow-origin information. This measure counteracts the imbalance between fast and slow agents (i.e. $\Delta T_s > \Delta T_f$).

This effect is modelled by modifying the probability of agent state change. Concretely, (9) shows that the outcome depends on the two cross-type probabilities only through the ratio P_{sf}/P_{fs} , scaled by the speed ratio. Holding $P_{fs} = p$ fixed and raising P_{sf} gives

$$\mathcal{C}(P_{sf}) = \frac{P_{sf}}{\rho p}, \quad (11)$$

which is linear in the intervention. Balance ($\mathcal{C} = 1$) is restored exactly when the effective cross-type transmission rates are equal,

$$r_s P_{sf} = r_f P_{fs} \iff P_{sf}^* = \rho P_{fs}. \quad (12)$$

For $p = 0.1$ this requires $P_{sf}^* = 0.1, 0.3, 1.0$ at $\rho = 1, 3, 10$. At $\rho = 100$ the required value $P_{sf}^* = 10$ exceeds unity: making fast agents maximally receptive to slow agents is no longer sufficient on its own, and balance additionally requires lowering P_{fs} (slow agents becoming less receptive to fast agents), e.g. $P_{sf} = 1, P_{fs} = 0.01$. The two cross-type probabilities are therefore the two levers of group alignment in this model; the within-type probabilities P_{ff} and P_{ss} have no effect on the long-run balance, only on the time (5) taken to reach it.

The experiment sweeps $P_{sf} \in [p, 1]$ at each ρ and tests whether the estimated $\mathcal{C}$ follows (11), crossing $\mathcal{C} = 1$ at P_{sf}^* . Compared with the uniform-probability case, slow-origin information is predicted to be represented P_{sf}/p times more strongly, so that at the balance point it competes on equal terms with fast-origin information.

The results from the experiment show that

[Placeholder – results pending]